

Experiment 1: Introduction to UNIX File System

Aim:

To study the structure of UNIX file system.

Theory:

UNIX file system is a hierarchical structure where all files and directories are organized in a tree-like structure starting from the root directory /.

Important Directories:

- / → Root directory
- /home → User files
- /bin → Essential commands
- /etc → Configuration files
- /tmp → Temporary files

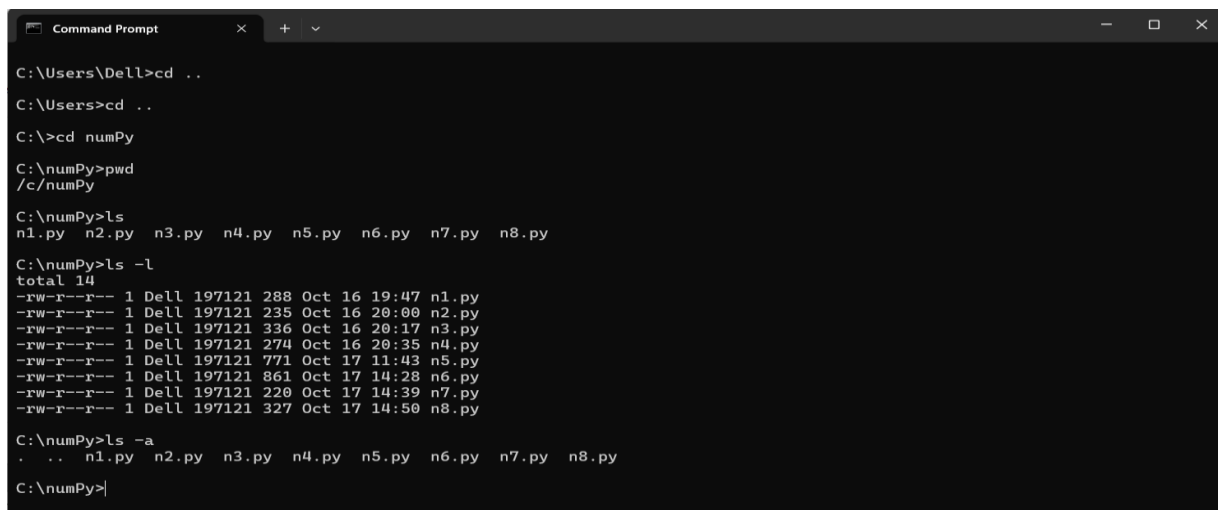
Commands:

```
pwd
ls
ls -l
ls -a
```

Procedure:

1. Open terminal.
2. Type `pwd` to check current directory.
3. Use `ls` to list files.
4. Use `ls -l` for detailed view.

Output:



```
Command Prompt
C:\Users\Dell>cd ..
C:\Users>cd ..
C:\>cd numPy
C:\numPy>pwd
/c/numPy
C:\numPy>ls
n1.py n2.py n3.py n4.py n5.py n6.py n7.py n8.py
C:\numPy>ls -l
total 14
-rw-r--r-- 1 Dell 197121 288 Oct 16 19:47 n1.py
-rw-r--r-- 1 Dell 197121 235 Oct 16 20:00 n2.py
-rw-r--r-- 1 Dell 197121 336 Oct 16 20:17 n3.py
-rw-r--r-- 1 Dell 197121 274 Oct 16 20:35 n4.py
-rw-r--r-- 1 Dell 197121 771 Oct 17 11:43 n5.py
-rw-r--r-- 1 Dell 197121 861 Oct 17 14:28 n6.py
-rw-r--r-- 1 Dell 197121 220 Oct 17 14:39 n7.py
-rw-r--r-- 1 Dell 197121 327 Oct 17 14:50 n8.py
C:\numPy>ls -a
. . n1.py n2.py n3.py n4.py n5.py n6.py n7.py n8.py
C:\numPy>
```

Experiment 2: File and Directory Related Commands

Aim:

To learn file and directory operations.

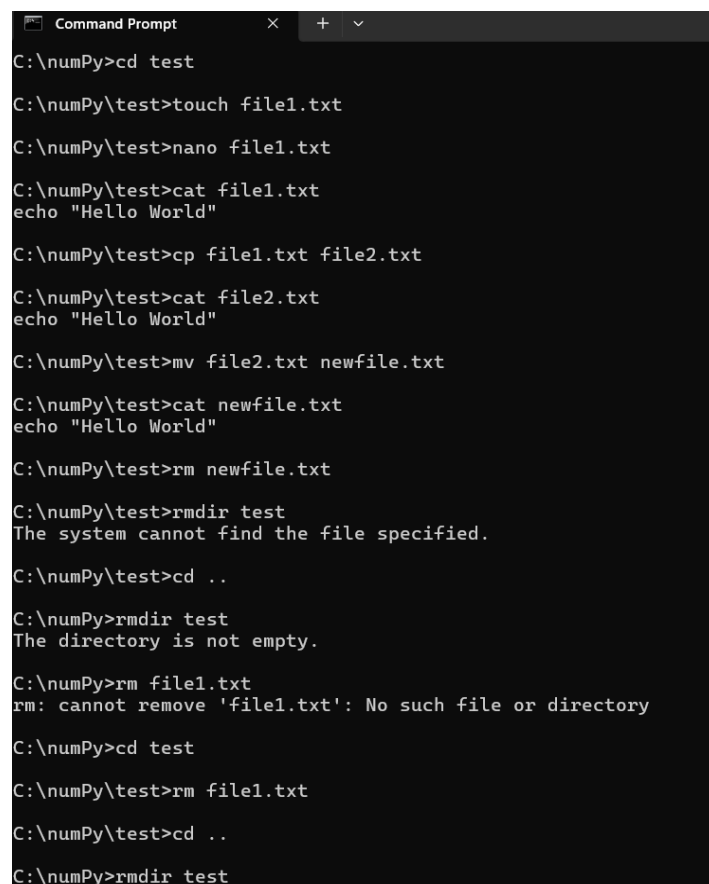
Commands:

```
mkdir test
cd test
touch file1.txt
cp file1.txt file2.txt
mv file2.txt newfile.txt
rm newfile.txt
rmdir test
```

Procedure:

1. Create directory using `mkdir`.
2. Navigate using `cd`.
3. Create file using `touch`.
4. Copy, move, and delete files.

Output:



```
Command Prompt
C:\numPy>cd test
C:\numPy\test>touch file1.txt
C:\numPy\test>nano file1.txt
C:\numPy\test>cat file1.txt
echo "Hello World"
C:\numPy\test>cp file1.txt file2.txt
C:\numPy\test>cat file2.txt
echo "Hello World"
C:\numPy\test>mv file2.txt newfile.txt
C:\numPy\test>cat newfile.txt
echo "Hello World"
C:\numPy\test>rm newfile.txt
C:\numPy\test>rmdir test
The system cannot find the file specified.
C:\numPy\test>cd ..
C:\numPy>rmdir test
The directory is not empty.
C:\numPy>rm file1.txt
rm: cannot remove 'file1.txt': No such file or directory
C:\numPy>cd test
C:\numPy\test>rm file1.txt
C:\numPy\test>cd ..
C:\numPy>rmdir test
```

Experiment 3: Essential UNIX Commands

Aim:

To study basic UNIX commands.

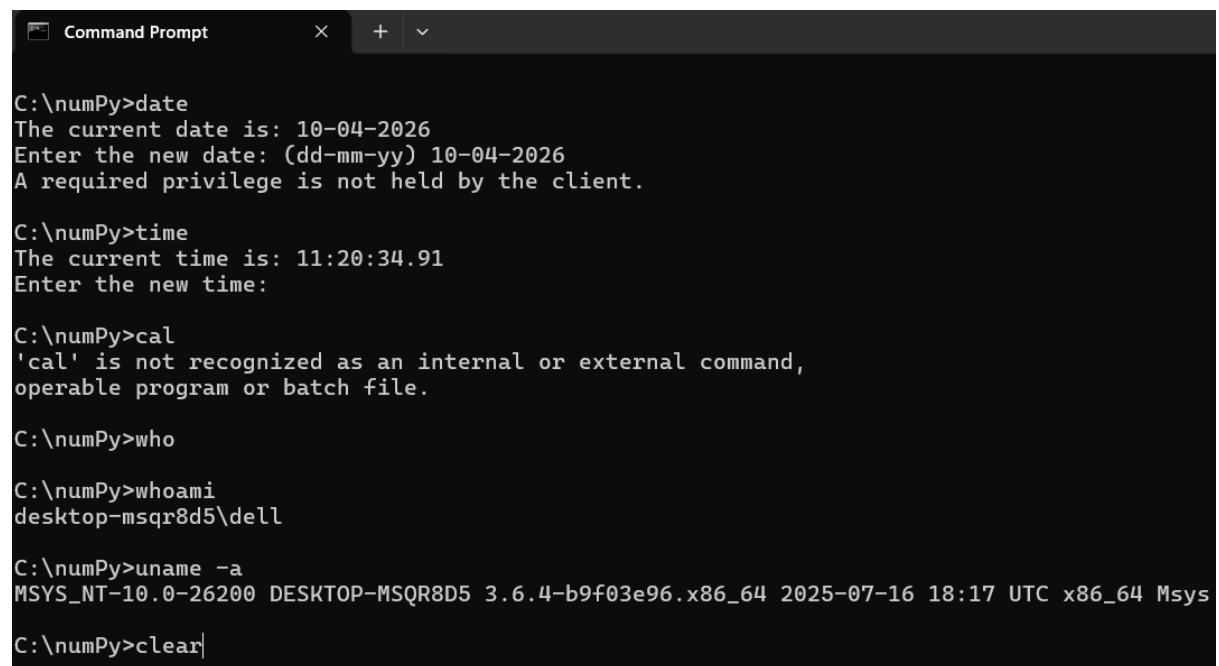
Commands:

```
date
cal
who
whoami
uname -a
clear
```

Procedure:

Execute each command in terminal and observe output.

Output:



```
Command Prompt
C:\numPy>date
The current date is: 10-04-2026
Enter the new date: (dd-mm-yy) 10-04-2026
A required privilege is not held by the client.

C:\numPy>time
The current time is: 11:20:34.91
Enter the new time:

C:\numPy>cal
'cal' is not recognized as an internal or external command,
operable program or batch file.

C:\numPy>who

C:\numPy>whoami
desktop-msqr8d5\dell

C:\numPy>uname -a
MSYS_NT-10.0-26200 DESKTOP-MSQR8D5 3.6.4-b9f03e96.x86_64 2025-07-16 18:17 UTC x86_64 Msys

C:\numPy>clear|
```

Experiment 4: I/O Redirection and Piping

Aim:

To understand input/output redirection.

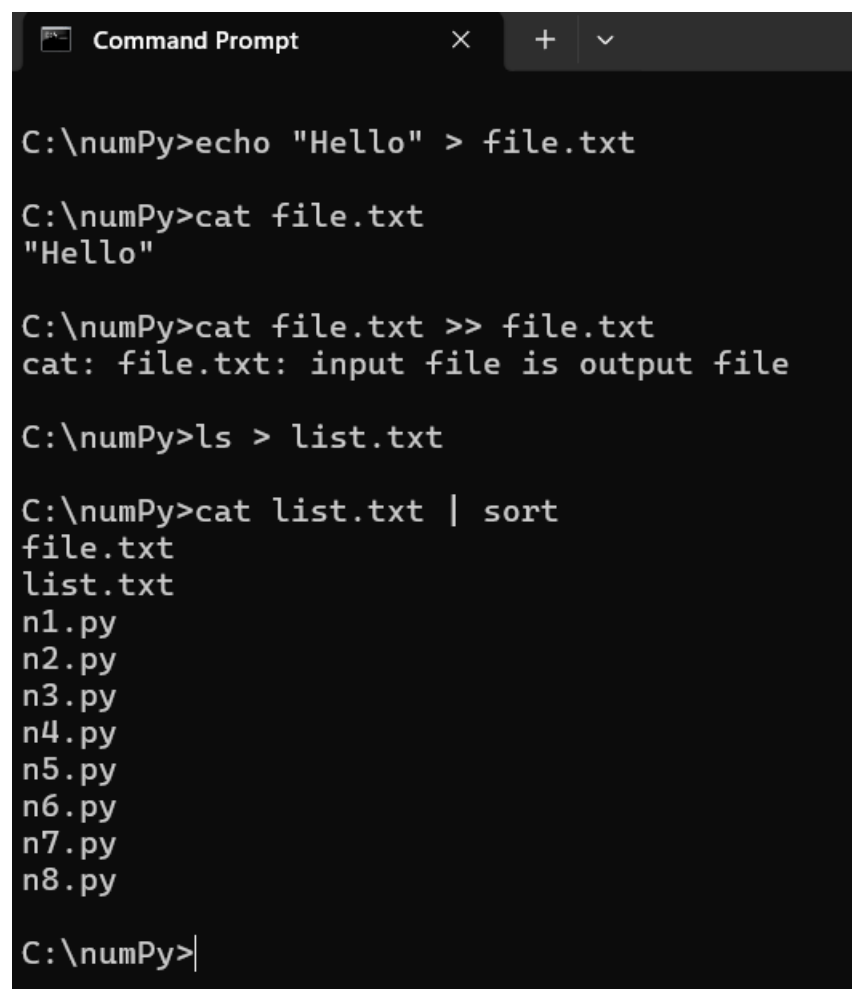
Commands:

```
echo "Hello" > file.txt
cat file.txt
cat file.txt >> file.txt
ls > list.txt
cat list.txt | sort
```

Explanation:

- > → overwrite
- >> → append
- | → pipe

Output:



```
Command Prompt

C:\numPy>echo "Hello" > file.txt

C:\numPy>cat file.txt
"Hello"

C:\numPy>cat file.txt >> file.txt
cat: file.txt: input file is output file

C:\numPy>ls > list.txt

C:\numPy>cat list.txt | sort
file.txt
list.txt
n1.py
n2.py
n3.py
n4.py
n5.py
n6.py
n7.py
n8.py

C:\numPy>
```

```
MINGW64/c/numPy
i "Hello"

file.txt [dos] (11:55 10/04/2026)
file.txt" [dos] 1L, 13B
```

Experiment 5: Introduction to VI Editor

Aim:

To learn VI editor basics.

Commands:

```
vi file.txt
```

Modes:

- Insert mode → press `i`
- Command mode → press `Esc`

Common Commands:

```
:w    (save)
:q    (quit)
:wq   (save & exit)
```

Output:

```
De11@DESKTOP-MSQR8D5 MINGW64 ~
$ cd ..

De11@DESKTOP-MSQR8D5 MINGW64 /c/Users
$ cd ..

De11@DESKTOP-MSQR8D5 MINGW64 /c
$ cd numpy

De11@DESKTOP-MSQR8D5 MINGW64 /c/numpy
$ vi file.txt
```

Experiment 6: Processes in UNIX

Aim:

To study process management.

Commands:

```
ps
top
kill <PID>
```

Procedure:

1. Run `ps` to view processes.
2. Use `top` for live processes.
3. Kill a process using `kill`.

Output:

```
MINGW64:/c/Users/Dell
Dell@DESKTOP-MSQR8D5 MINGW64 ~
$ ps
  PID   PPID  PGID   WINPID   TTY      UID         STIME  COMMAND
 1164     1  1164   13660    ?        197609  12:03:15 /usr/bin/mintt
 1165   1164  1165   16744  pty0      197609  12:03:15 /usr/bin/bash
 1187   1165  1187   2808    pty0      197609  12:03:22 /usr/bin/ps

Dell@DESKTOP-MSQR8D5 MINGW64 ~
$ tasklist

Image Name                      PID Session Name        Session#    Mem Usage
-----
System Idle Process              0 Services              0             8 K
System                           4 Services              0          2,896 K
Secure System                   116 Services              0         39,432 K
Registry                        160 Services              0         58,348 K
smss.exe                        648 Services              0             12 K
csrss.exe                       368 Services              0          3,352 K
wininit.exe                     440 Services              0          1,724 K
csrss.exe                       908 Console                1          3,392 K
winlogon.exe                   1076 Console                1          2,660 K
services.exe                   1148 Services              0          9,868 K
lsass.exe                      1168 Services              0          1,328 K
lsass.exe                      1180 Services              0         17,816 K
svchost.exe                     1308 Services              0         23,380 K
fontdrvhost.exe               1320 Console                1          2,100 K
fontdrvhost.exe               1328 Services              0          1,600 K
WUDFHost.exe                   1408 Services              0          2,908 K
svchost.exe                     1472 Services              0         13,968 K
svchost.exe                     1532 Services              0          4,416 K
WUDFHost.exe                   1572 Services              0          8,448 K
dwm.exe                        1664 Console                1        1,25,428 K
svchost.exe                     1756 Services              0          1,916 K
svchost.exe                     1764 Services              0          4,052 K
svchost.exe                     1796 Services              0          4,368 K
WUDFHost.exe                   1812 Services              0          5,548 K
svchost.exe                     1896 Services              0          3,184 K
svchost.exe                     1940 Services              0          5,772 K
svchost.exe                     1952 Services              0          4,092 K
svchost.exe                     2008 Services              0          5,228 K
IntelCpHDCPSvc.exe            1528 Services              0          1,772 K
svchost.exe                     1652 Services              0          2,620 K
svchost.exe                     1888 Services              0          3,132 K
svchost.exe                     1492 Services              0          9,328 K
svchost.exe                     2068 Services              0          1,908 K
svchost.exe                     2300 Services              0          6,432 K
svchost.exe                     2332 Services              0          5,304 K
svchost.exe                     2376 Services              0          2,244 K
svchost.exe                     2584 Services              0          8,460 K
svchost.exe                     2672 Services              0          4,348 K
svchost.exe                     2792 Services              0         10,112 K
svchost.exe                     2880 Services              0         11,764 K
svchost.exe                     2928 Services              0          6,412 K
svchost.exe                     2936 Services              0          2,316 K
svchost.exe                     2952 Services              0          6,656 K
Memory Compression             2176 Services              0         96,160 K
svchost.exe                     2204 Services              0          3,464 K

Dell@DESKTOP-MSQR8D5 MINGW64 ~
$ kill 1187
```

Experiment 7: Communication in UNIX & AWK

Aim:

To study communication and AWK tool.

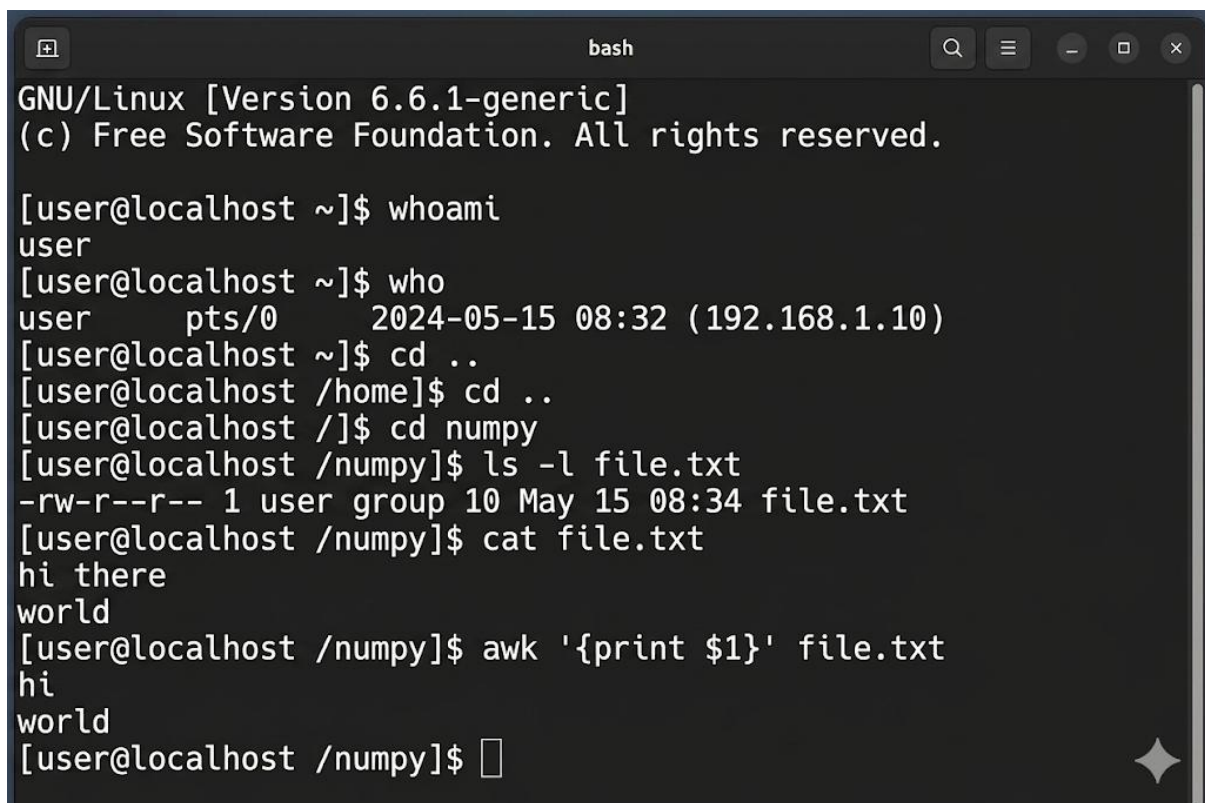
Commands:

```
who  
write username  
awk '{print $1}' file.txt
```

Explanation:

AWK is used for text processing.

Output:

A terminal window titled 'bash' with standard window controls. It shows the execution of several commands: 'whoami' returns 'user'; 'who' shows user details; 'cd ..' is executed twice to move from the home directory to the root; 'ls -l file.txt' shows file permissions and details; 'cat file.txt' displays the file content 'hi there' and 'world'; and 'awk '{print \$1}' file.txt' prints 'hi' and 'world' on separate lines.

```
GNU/Linux [Version 6.6.1-generic]  
(c) Free Software Foundation. All rights reserved.  
  
[user@localhost ~]$ whoami  
user  
[user@localhost ~]$ who  
user      pts/0      2024-05-15 08:32 (192.168.1.10)  
[user@localhost ~]$ cd ..  
[user@localhost /home]$ cd ..  
[user@localhost /]$ cd numpy  
[user@localhost /numpy]$ ls -l file.txt  
-rw-r--r-- 1 user group 10 May 15 08:34 file.txt  
[user@localhost /numpy]$ cat file.txt  
hi there  
world  
[user@localhost /numpy]$ awk '{print $1}' file.txt  
hi  
world  
[user@localhost /numpy]$
```

Experiment 8: Shell Scripting Introduction

Aim:

To understand shell scripting.

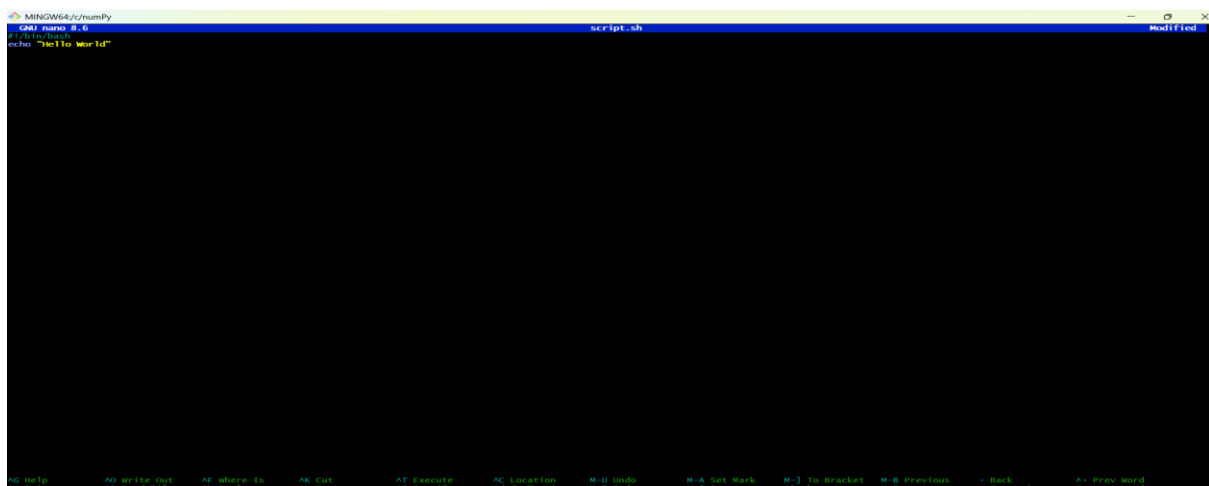
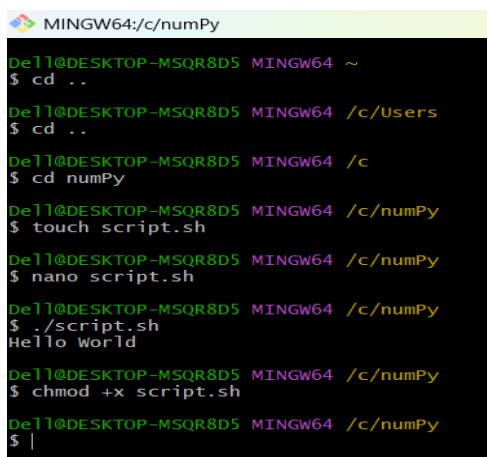
Example Script:

```
#!/bin/bash
echo "Hello World"
```

Run Command:

```
chmod +x script.sh
./script.sh
```

Output:

A screenshot of a terminal window titled 'MINGW64/c/numPy'. The window shows the execution of a script named 'script.sh'. The prompt is 'MINGW64 ~', and the user has entered 'cd ..', 'cd /c/Users', 'cd ..', 'cd /c', 'cd numPy', 'touch script.sh', 'nano script.sh', './script.sh', and 'chmod +x script.sh'. The output of the script is 'Hello World'.A screenshot of a terminal window titled 'MINGW64/c/numPy'. The window shows the steps to create and run a shell script. The prompt is 'De11@DESKTOP-MSQR8D5 MINGW64 ~'. The user enters the following commands: 'cd ..', 'cd /c/Users', 'cd ..', 'cd /c', 'cd numPy', 'touch script.sh', 'nano script.sh', './script.sh', and 'chmod +x script.sh'. The output of the script is 'Hello World'.

Experiment 9: Decision & Iteration in Shell

Aim:

To implement decision and loops.

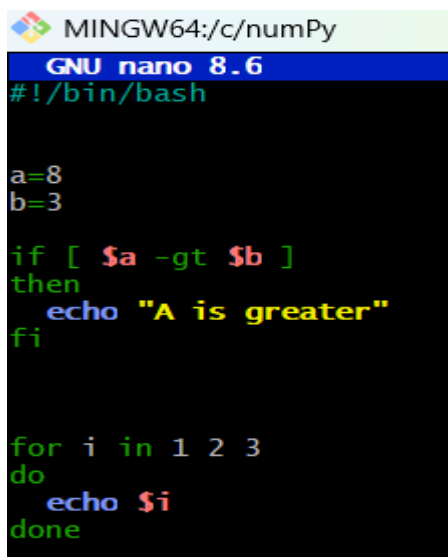
If Statement:

```
if [ $a -gt $b ]
then
echo "A is greater"
fi
```

Loop:

```
for i in 1 2 3
do
echo $i
done
```

Output:

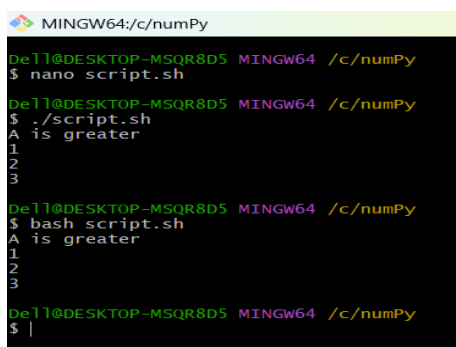


```
MINGW64:/c/numPy
GNU nano 8.6
#!/bin/bash

a=8
b=3

if [ $a -gt $b ]
then
    echo "A is greater"
fi

for i in 1 2 3
do
    echo $i
done
```



```
MINGW64:/c/numPy
Dell@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ nano script.sh
Dell@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ ./script.sh
A is greater
1
2
3
Dell@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ bash script.sh
A is greater
1
2
3
Dell@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ |
```

Experiment 10: Shell Scripts for Problems

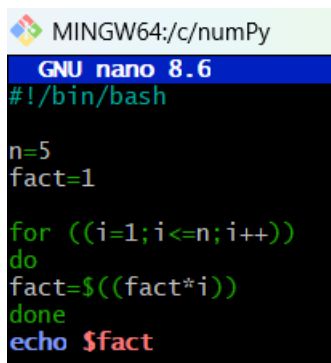
Aim:

To solve problems using scripts.

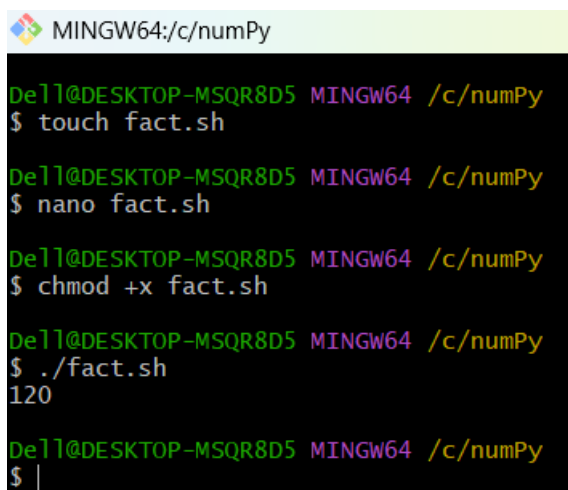
Example: Factorial Program

```
#!/bin/bash
n=5
fact=1
for ((i=1;i<=n;i++))
do
fact=$((fact*i))
done
echo $fact
```

Output:

A screenshot of a terminal window with a light green title bar that reads 'MINGW64:/c/numPy'. The terminal content shows the GNU nano 8.6 editor interface. The script content is being edited:

```
#!/bin/bash
n=5
fact=1
for ((i=1;i<=n;i++))
do
fact=$((fact*i))
done
echo $fact
```

A screenshot of a terminal window with a light green title bar that reads 'MINGW64:/c/numPy'. The terminal shows the following sequence of commands and output:

```
De11@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ touch fact.sh

De11@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ nano fact.sh

De11@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ chmod +x fact.sh

De11@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ ./fact.sh
120

De11@DESKTOP-MSQR8D5 MINGW64 /c/numPy
$ |
```